Math-M-Addicts
MINDS THAT COUNT

SP250 Homework 5

1. Find the smallest possible positive integer which has all 10 digits in its decimal representation and is divisible by 72.

Solution:

A number is divisible by 72 if it is divisible by both 8 and 9.

The sum of all 10 digits is 0+1+2+...+8+9=45, which is divisible by 9. By the divisibility rule for 9, any positive integer consisting of all 10 digits will be divisible by 9. Thus, the smallest number consisting of all 10 digits that is divisible by 9 is just 1,023,456,789, where the digits, except for 0, are arranged from smallest to largest from left to right. The digit 0 is an exception because a number cannot start with 0.

Now, a number is divisible by 8 if its last 3-digits is divisible by 8 (because 1000 is divisible by 8), so we just need to switch around some of the digits of 1,023,456,789 to make its last 3-digits divisible by 8. First, we try to make a 3-digit number using {7,8,9} that's divisible by 8. Since anything divisible by 8 has to end with an even digit and neither 798 nor 978 is divisible by 8, this is not possible. Next, we try using the digits {6,8,9}, where the digit 7, the smallest among 7,8,9, is replaced with 6, the largest among the other digits (so to keep the first 7-digits as small as possible). We find that 896 and 968 are both divisible by 8, so we go with the smaller 896. Thus, the smallest number that contains all 10 digits and is divisible by 72 is **1,023,457,896**.

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2. Heracles needs to fight the Lernean Hydra for one of his labors. A Hydra has many heads, and to defeat it, he must remove them all.

Heracles has two swords, and he can only use one of them at a time. The smaller sword will sever 21 of the Hydra's heads with a single stroke. The larger one will sever 100 heads with each stroke, but then the Hydra (unless it has just been killed) will grow 2023 new heads! If the Hydra initially has 101 heads, is it possible for Heracles to defeat it with his present swords or does he need to procure another sword?

NOTE: *A sword can only remove a particular number of heads, no fewer and no more.* If, for example, the Hydra has just 5 heads, they cannot be cut off with either of the swords.

Solution:

In order for Heracles to kill the Hydra, the number of heads on the Hydra just before the final killing blow must be either 21 or 100.

Every time Heracles uses the large sword without killing the Hydra, the number of heads on the Hydra increases by $2023 - 100 = 1923$.

Now, let x and y be the number of times Heracles used the small and big swords respectively before the

final killing blow. Then, in order for Heracles to kill the hydra, there must exist integers x, y such that

$$101 - 21x + 1923y = 21 \text{ or } 100$$

Now, let's consider the left and right hand sides of the above equation in modulo 3.

For the left-hand side, we have $101 - 21x + 1923y \equiv 2 \ (mod\ 3)$ as 21 and 1923 are both multiples of 3 and $101 \equiv 2 \ (mod\ 3)$.

For the right-hand side, we have either $21 \equiv 0 \ (mod\ 3)$ or $100 \equiv 1 \ (mod\ 3)$.

So the left and right hand sides of the equation can never be equal for any integers x, y. Thus, it is impossible for Heracles to kill the Hydra with the two swords that he has.

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3. Which number is larger: 2^{700} or 5^{300}?

Solution:

$2^{700} = (2^7)^{100} = 128^{100}$

$5^{300} = (5^3)^{100} = 125^{100}$

Since $128 > 125$, we have $2^{700} = 128^{100} > 125^{100} = 5^{300}$.

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4. The number of students in a certain classroom is between 30 and 40. It is known that each girl shook hands with exactly 3 boys but each boy shook hands with exactly 4 girls. How many students might be present in the classroom? Prove the uniqueness of your answer!

Solution:

Let there be g girls and b boys. The girls all together shook hands with boys $3g$ times and the boys all together shook hands with girls $4b$ times, so we have $3g = 4b$. This implies $g = \frac{4}{3}b$ and the total number of students is $b + g = b + \frac{4}{3}b = \frac{7}{3}b = 7(\frac{b}{3})$. This means that the total number of students is a multiple of 7. The only multiple of 7 between 30 and 40 is 35, so there were 35 students in the classroom.

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5. Prove that the only integer solution to the equation $3 \cdot x^2 = y^2 + z^2$ is $x = y = z = 0$.

Solution:

The right hand side of the equation, $3x^2$, is 0 mod 3. Since squares can only be 0 or 1 mod 3, both y^2 and z^2 must be 0 mod 3, so y and z are both divisible by 3. Let $y = 3v$ and $z = 3w$ for some integers v and w. Then we have

$$3x^2 = 9v^2 + 9w^2 \quad \Rightarrow \quad x^2 = 3v^2 + 3w^2$$

Which means x^2 is also 0 mod 3 and x is also divisible by 3.

Now let's assume there are nonzero integer solutions to the equation. For the solution x, y, z with the smallest possible positive x value, we have $x = 3u, y = 3v, z = 3w$ for some integers u, v, w since x, y, z are all divisible by 3. Now, we have

$$3x^2 = y^2 + z^2 \quad \Rightarrow \quad 3(9u^2) = 9v^2 + 9w^2 \quad \Rightarrow \quad 3u^2 = v^2 + w^2$$

We see that u, v, w also satisfies the same equation, which is a contradiction since u is a positive integer smaller than x but x was supposed to be the smallest. Hence, our assumption must have been false and the only possible solution is $x = y = z = 0$.

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6. There are 7 rooks on a 7×7 board so that none attacks another. Show that any 4×4 square contains at least 1 rook.

Solution:
With 7 rooks on a 7 x 7 board with none attacking each other, there must be exactly one rook in each row as well as exactly one rook in each column.

Now, let's consider any 4 consecutive rows on the board. The 4 consecutive rows form a 4 x 7 rectangle. There are exactly 4 rooks in these 4 consecutive rows as there is exactly one rook in each row. These 4 rooks must sit in 4 different columns, so there are only 3 columns of the 4 x 7 rectangle that are not occupied by one of these 4 rooks. Therefore, any 4 x 4 square must contain at least one rook.
